ATHARV SHEERSH CHATURVEDI

Bilaspur, Chhattisgarh, India | 9039860092 | atharvsc26@gmail.com | GitHub | LinkedIn

EDUCATION

Kalinga Institute of Industrial Technology, Bhubaneswar, Odisha July 2023 – Present
Bachelor of Technology (B.Tech) | CGPA: 7.9 / 10.0

TECHNICAL SKILLS

Programming: Python, Embedded C, HTML, CSS, JavaScript
Tools & Frameworks: Flask, Matplotlib, Pygame, REST APIs
Core Areas: Backend Development, AI, Data Visualization, IoT, Game Development

WORK EXPERIENCE

Embedded Engineer — KIIT Robotics Society, Bhubaneswar Nov 2024 – Jun 2026

- Educated 50+ students on IoT and Embedded Systems through KRS Classes, represented KRS in KSHITIJ 2025 and KREATIVITY 2.0.
- Delivered technical presentations on ESP32 applications, sensor integration, and wireless communication protocols.

IoT Engineer — IoT Lab, Bhubaneswar Sep 2025 – Jan 2026

- Developed object detection systems using computer vision and machine learning for surveillance applications.
- Implementing IoT sensor networks for real-time environmental monitoring with edge computing solutions.

PROJECTS

DocuStream AI — Python, Flask, Chroma, Gemini, Groq Live · GitHub
RAG-based research assistant with page-cited Q&A and an **LLM**-scored paper **Readiness Score** against real reviewer criteria; built on Gemini embeddings, Chroma vector search, and Groq inference, deployed on Vercel.

Campus-Navigator — Python, Flask, A* Search, Geospatial Reasoning, REST API, SMTP Live · GitHub
Built an **Agentic AI**-powered campus navigation system using **A* search** with geospatial heuristics for autonomous pathfinding across a real-world weighted road graph. Developed a full-stack Flask REST API with real-time path visualization, dynamic map rendering, and an automated SMTP feedback pipeline.

Genetic Algorithm for Ballistic Missile Trajectory Optimization — Python, NumPy, Matplotlib, Genetic Algorithm GitHub
Engineered a **path optimization** system using a custom **Genetic Algorithm** (population: 200, up to 500 generations) to compute fuel-efficient, obstacle-avoiding routes through a 3D constrained environment with dynamic no-go zones. Implemented a weighted multi-objective fitness function (path cost + directional penalty + constraint violations) with elitism (10%), tournament selection, Gaussian mutation, and Lamarckian repair — achieving convergence within ~300 generations at sub-3-unit target accuracy.

CERTIFICATIONS & ACHIEVEMENTS

- Google Cloud — Google Cloud Fundamentals: Core Infrastructure
- Cisco AICTE — Virtual Internship Program
- Amazon Web Services (AWS) — AWS Academy Graduate - Cloud Foundations
- Python — Basic to Advanced | Data Science and ML Bootcamp
- ExcelR Edtech Pvt. Ltd. with KIIT University — SAP BTP Developer
- IIM Ahmedabad — Strategy and Game Theory for Management
- University of Colorado — Business Analytics for Decision Making
- TCS iON Career Edge — Young Professional, IT Primer, Interview & Job Readiness
- Smart India Hackathon — Participant | KREATIVITY 2.0 Project Expo — Prototyping & Demonstration
- ServiceNow University, AICTE & SmartBridge — Virtual Internship Program
- 3rd National Conclave on Viksit Bharat @2047 — Paper Presentation | IIT-KGP Bhubaneswar Tech Workshop — Certificate of Participation

ABOUT

Passionate AI developer exploring IoT and game development. Creative writer (pen name: Abhay Pathak). Enthusiast of Cinema, Cricket, and Formula 1. Languages: Hindi (Native), English (Fluent).